50Days-coding-challenge

Day-16[SQL]

Customer Segmentation

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Questions

Level 1: Basic Filtering

1. Show all customers from Bangalore

Query | Query History | Scratch Pad

```sql
customer_name VARCHAR(50),
city VARCHAR(50),
age INT,
total_spent DECIMAL(10,2),
number_of_orders INT );

INSERT INTO customers VALUES (1, 'Amit', 'Bangalore', 25, 12000, 5), (2, 'Neha',

select* from customers;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	customer_id integer	customer_name character varying (50)	city character varying (50)	age integer	total_spent numeric (10,2)	number_of_orders integer
1	1	Amit	Bangalore	25	12000.00	5
2	4	Sneha	Bangalore	28	60000.00	15
3	7	Arjun	Bangalore	40	70000.00	20

Total rows: 3 | Query complete 00:00:00.243 | CRLF | Ln 13, Col 24

2. Display customers with total_spent > 20000

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```
INSERT INTO customers VALUES (1, 'Amit', 'Bangalore', 25, 12000, 5), (2, 'Neha',

select* from customers;

select*from customers
where city='Bangalore';

select*from customers
where total_spent>20000;
```

Showing rows: 1 to 3 Page No: 1 of 1

	customer_id integer	customer_name character varying (50)	city character varying (50)	age integer	total_spent numeric (10,2)	number_of_orders integer
1	2	Neha	Mumbai	30	45000.00	12
2	4	Sneha	Bangalore	28	60000.00	15
3	7	Arjun	Bangalore	40	70000.00	20

Total rows: 3 Query complete 00:00:00.485 CRLF Ln 15, Col 1

3. List customers aged between 25 and 35

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```
select* from customers;

select*from customers
where city='Bangalore';

select*from customers
where total_spent>20000;

select * from customers
where age between 25 and 35;
```

Showing rows: 1 to 5 Page No: 1 of 1

	customer_id integer	customer_name character varying (50)	city character varying (50)	age integer	total_spent numeric (10,2)	number_of_orders integer
1	1	Amit	Bangalore	25	12000.00	5
2	2	Neha	Mumbai	30	45000.00	12
3	4	Sneha	Bangalore	28	60000.00	15
4	5	Karan	Mumbai	35	20000.00	7
5	6	Pooja	Delhi	27	15000.00	6

Total rows: 5 Query complete 00:00:00.600 CRLF Ln 18, Col 1

Level 2: Sorting & Aggregation

1. Show all customers sorted by total_spent (highest first)

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Query | Query History

```sql
CREATE TABLE customers ( customer_id INT,
customer_name VARCHAR(50),
city VARCHAR(50),
age INT,
total_spent DECIMAL(10,2),
number_of_orders INT );
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 Page No: 1 of 1

	customer_id integer	customer_name character varying (50)	city character varying (50)	age integer	total_spent numeric (10,2)	number_of_orders integer
1	7	Arjun	Bangalore	40	70000.00	20
2	4	Sneha	Bangalore	28	60000.00	15
3	2	Neha	Mumbai	30	45000.00	12
4	5	Karan	Mumbai	35	20000.00	7
5	6	Pooja	Delhi	27	15000.00	6
6	1	Amit	Bangalore	25	12000.00	5
7	3	Raj	Delhi	22	8000.00	3
8	8	Meera	Mumbai	23	5000.00	2

Total rows: 8 | Query complete 00:00:00.246 | CRLF | Ln 21, Col 1

2. Find the total revenue generated

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Query | Query History

```sql
where age between 25 and 35;

select*from customers
order by total_spent desc;

select sum(total_spent) as Total_Revenue from customers;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	total_revenue numeric
1	235000.00

Total rows: 1 | Query complete 00:00:00.290 | CRLF | Ln 24, Col 1

3. Find the average spending per customer

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Query | Query History

```
select*from customers
order by total_spent desc;

select sum(total_spent) as Total_Revenue from customers;

select avg(total_spent) as Avg_spent_per_customer from customers;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	avg_spent_per_customer numeric
1	29375.000000000000

Total rows: 1 | Query complete 00:00:00.180 | CRLF | Ln 26, Col 66

Level 3: Grouping

1. Find total spending per **city**

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Query | Query History

```
select sum(total_spent) as Total_Revenue from customers;

select avg(total_spent) as Avg_spent_per_customer from customers;

select city,sum(total_spent) as Total_spend_per_city from customers
group by city;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	city character varying (50)	total_spend_per_city numeric
1	Mumbai	70000.00
2	Delhi	23000.00
3	Bangalore	142000.00

Total rows: 3 | Query complete 00:00:00.211 | CRLF | Ln 29, Col 15

2. Count number of customers in each **city**

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Query | Query History

```
select city,sum(total_spent) as Total_spend_per_city from customers
group by city;

select city,count(customer_id) as No_of_customers from customers
group by city;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	city character varying (50)	no_of_customers bigint
1	Mumbai	3
2	Delhi	2
3	Bangalore	3

Total rows: 3 | Query complete 00:00:00.237 | CRLF | Ln 31, Col 1

Level 4: Customer Segmentation (CORE TASK)

9.Count how many customers fall into each segment

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Query | Query History

```
select
case
when total_spent>50000 then 'High spent'
when total_spent between 20000 and 50000 then 'Medium spent'
else 'Low spent' end as customer_segment,
count(*) as total_customers from customers

group by customer_segment
order by Total_customers desc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	customer_segment text	total_customers bigint
1	Low spent	4
2	Medium spent	2
3	High spent	2

Total rows: 3 | Query complete 00:00:00.343 | CRLF | Ln 42, Col 31

Level 5: Advanced Basic (HAVING)

10.Show cities where total spending is **greater than 50000**

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No limit

Query | Query History

```sql
when total_spent>50000 then 'High spent'
when total_spent between 20000 and 50000 then 'Medium spent'
else 'Low spent' end as customer_segment,
count(*) as total_customers from customers

group by customer_segment
order by Total_customers desc;

select city,sum(total_spent) as Total_spending from customers
group by city
having sum(total_spent)>50000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

	city character varying (50)	total_spending numeric
1	Mumbai	70000.00
2	Bangalore	142000.00

Total rows: 2 | Query complete 00:00:00.255 | CRLF | Ln 44, Col 1